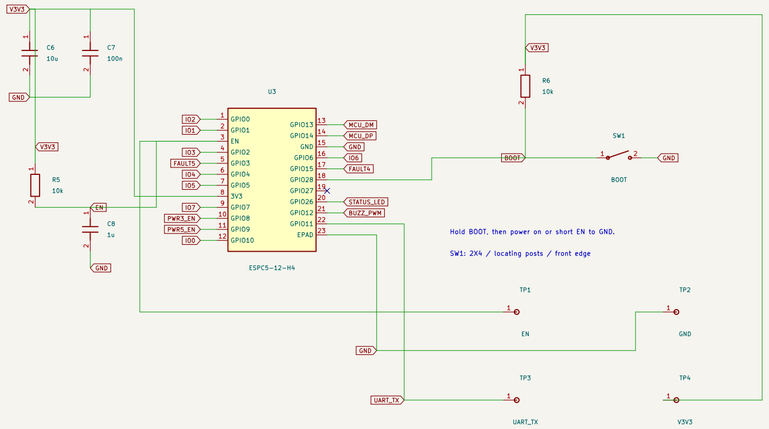
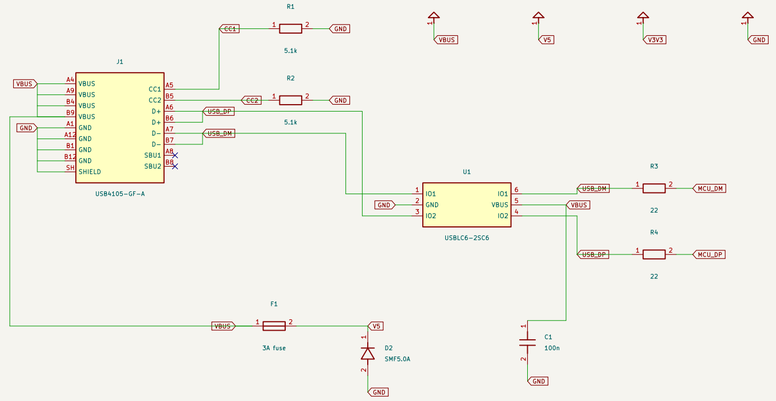
01 CONTROLLER / BOOT / RESET



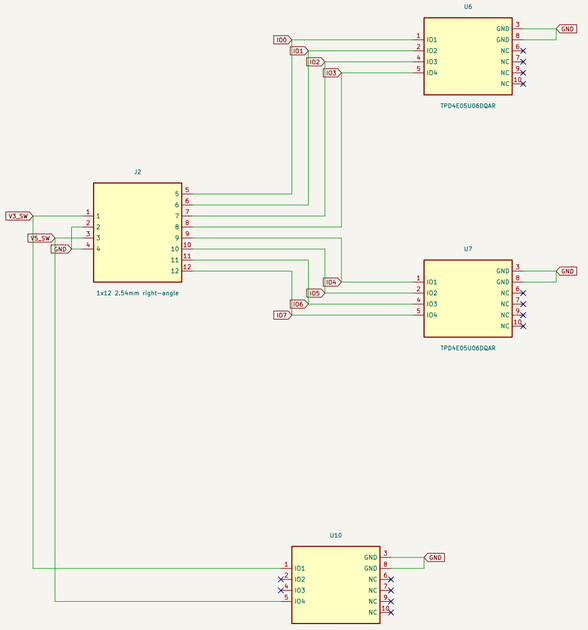
Built-in PCB antenna over cutout; 3.3 V pins only.

02 USB-C / INPUT PROTECTION



5 V / 3 A supply. No PD or OTG. External loads OFF on PC USB.

06 HEADER / ESD / SHUNT PROTECTION



U6/U7: four channels each; no supply or bypass needed.

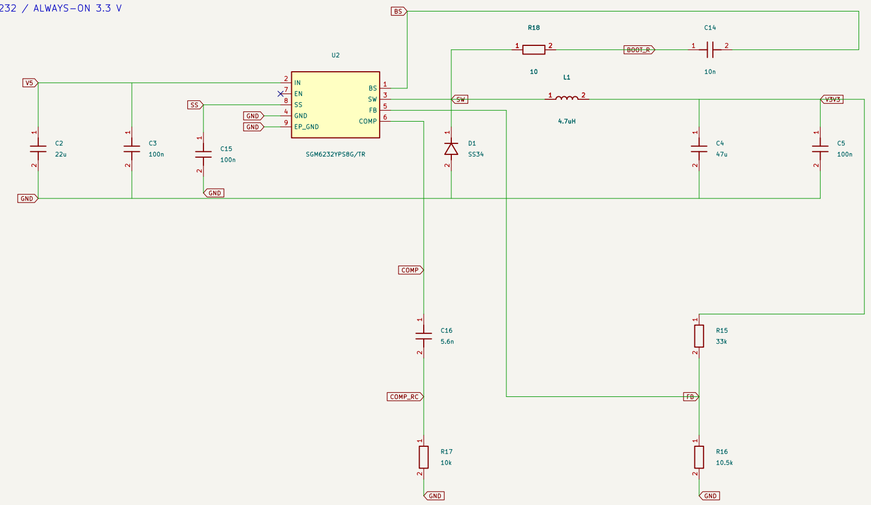
IO0–IO7: 3.3 V only; final-board ESD test required.

U10: switched rails. Power current bypasses the array.

Place protection arrays directly beside J2.

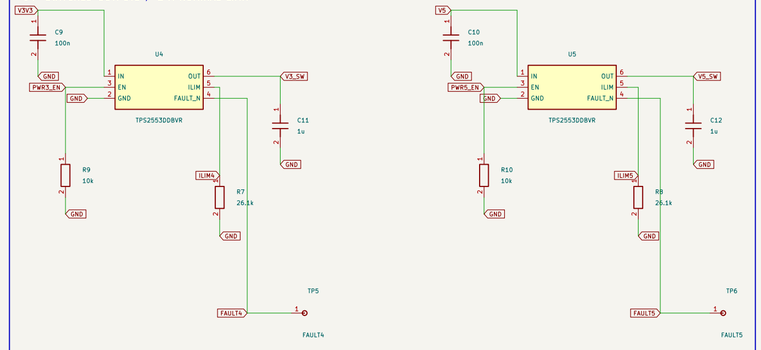
Short, wide ground returns; qualify ESD on final PCB.

03 SGM6232 / ALWAYS-ON 3.3 V



EN open: auto-start. EP to GND. FB divider 33k / 10.5k = 3.314 V.

04 SWITCHED OUTPUTS / 1 A NOMINAL LIMIT



Fault inputs: firmware must enable internal pullups.

26.1k ILIM: nominal 0.989 A; bounds 0.908–1.081 A.

FAULT4 -> input 15; FAULT5 -> input 3. Active LOW.

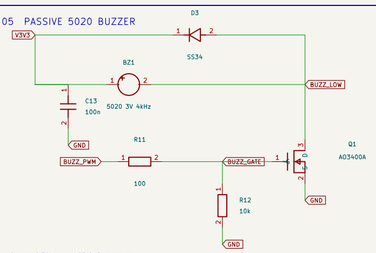
PROTOTYPE VALIDATION

Check startup, dropout, ripple, heat.

Use specified 5 V / 3 A adapter.

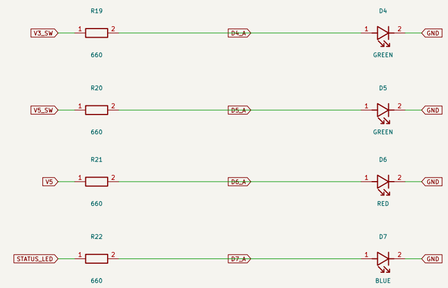
Verify current limit and both loaded rails.

05 PASSIVE 5020 BUZZER



4 kHz / 50% duty. GPIO12 reserved.

07 GROUPED STATUS INDICATORS



Blue: control pin 26, HIGH = on. 3V3/5V: switched rails.

48 x 28 mm PCB — all components on front surface.

48 x 28 mm / 2 layers / SGM6232 / prototype

GPT6-Astra / xqe2011

Sheet: /

File: saihub.kicad_sch

Title: SAIHub-Mini — IO CONTROLLER

Size: A2	Date: 2026-09-24	**Rev: A**
KiCad E.D.A. 10.0.6		Id: 1/1